

10 BILLION DOLLARS

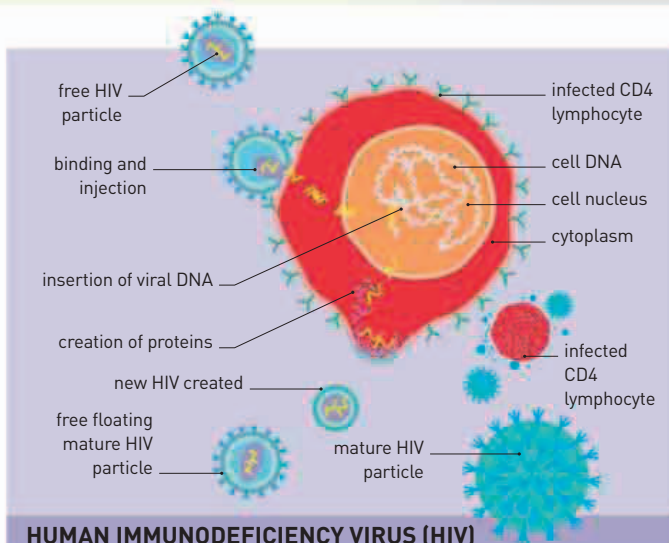
THE COST OF DAMAGE CAUSED BY EL NIÑO IN 1982-83

In 1982, El Niño caused sustained high rainfall, resulting in the raised water level in the San Lorenzo River, California.

AN IMPORTANT DEVELOPMENT IN DIGITAL SOUND reproduction, the **compact disk (CD)**, became available in October this year.

The CD, developed jointly by Philips and Sony, is a polycarbonate disk with a thin layer of aluminum inside. The aluminum is pitted with **millions of tiny indentations**, arranged as a spiral track more than 3 miles (5km) long. The pits represent binary digits, which in turn represent the original sound. A laser inside the CD player reflects light off the pits as the disk spins, and a microprocessor reconstructs the original sound wave from the pattern of reflected light. Within a few years, **CDs became the most popular format** for buying recorded music. The compact disk was adapted as a medium for read-only computer data storage (CD-ROM), and later, as writable data disks (CDRs).

Also in October, American synthesizer pioneer **Robert Moog** announced the **Musical Instrument Digital Interface (MIDI)**, a new way of recording and playing back musical performances. MIDI consists of simple messages that relate to the notes played; the messages can be produced by playing MIDI instruments such as keyboards, or by manipulating software. MIDI messages trigger sound



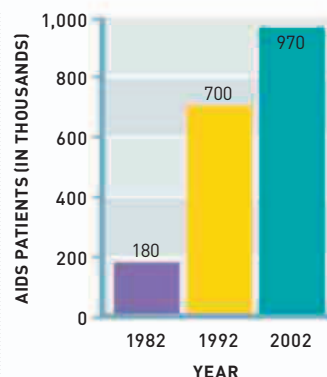
The cause of AIDS was identified as a virus in 1984, and that virus was given the name human immunodeficiency virus (HIV) in 1986. HIV is transmitted in bodily fluids and infects cells crucial to the body's immune system—in particular, CD4 cells—and uses them to reproduce. The cells may die as a result or be destroyed by other immune system cells.

samples, giving musicians great flexibility in composition, recording, and performance.

In July, a recently identified disease was given its name: **acquired immunodeficiency syndrome (AIDS)**. The disease had claimed many lives in the gay community in New York and California. However, it had become clear that, while the disease spread easily among homosexual men via sexual contact, it was by no means

confined to the gay community. The **rapid spread of the disease** led to major health campaigns, encouraging people to use condoms during intercourse, and intravenous drug users to avoid sharing needles.

Widespread extreme weather alerted the general public to the **existence of the El Niño phenomenon**—in which the Pacific Ocean remains warmer for longer than usual because of a change in the world's trade



Living with AIDS

This graph shows the number of people living with AIDS in the USA. Worldwide, 30 million people were living with the disease by 2002.

winds (the prevailing easterly winds found in the tropics). El Niño events happen occasionally, and typically last for several months. This year's El Niño, **one of the most devastating on record**, began in July and lasted into the next year. It decimated fish stocks in Peru, brought drought and bush fires to Australia and parts of Africa, and extreme rainfall in California and Peru—causing an estimated 2,000 deaths.

Artificial heart

A Jarvik-7 artificial heart like this was given to Barney Clark at the University of Utah. Clark lived for 112 days after the operation.

In May, Taiwanese biologist **Chiaho Shih** (b.1950) and American biologist **Robert Weinberg** (b.1942) reported that they had isolated a human oncogene, which is a **genetic cause of cancer**.

In December, American surgeon William DeVries (b.1943) performed surgery on retired dentist Barney Clark to implant the **first permanent artificial human heart**, the Jarvik-7. The artificial heart, designed by American inventor Robert Jarvik (b.1946), kept Clark alive for 112 days.



January US computer company Commodore introduces the hugely successful Commodore 64 personal computer

February 1 Intel introduces the first microprocessor with more than one hundred thousand transistors on it

April 9 American neurologist Stanley Prusiner defines the term "prions" to describe the newly discovered infectious particles that cause scrapie, BSE, and CJD

May 1 Isolation of first human oncogene (cancer-causing gene) announced

July Onset of weather phenomenon known as El Niño

July 27 At a meeting in Washington DC, the disease now known as AIDS gains its name

October 1 Sony launches the world's first commercially released compact disk player, the Sony CDP-101